

25 Z transform cont. cont.

Online lecture.

Questions

Why is Z a circle instead of a half plane?

Remember the definition of the z transform. The magnitude of the thing has a restriction, which keeps it in a radius.

For the Laplace transform, the real part has the restriction - not the magnitude. This is why the convergence areas are planes instead of circular radii.

What does z^{-1} mean in the real world?

We saw last time that different powers of z correspond to time domain signals (see the polynomial expansion). The coefficient for z^{-1} is the value for one time step in the future (a unit delay).

Region of Convergence

Multiplying a signal by something unstable that blows up would make the region of convergence shrink, or the opposite - if you multiply by a stable thing, the region of convergence would expand.

In the real world, we're dealing with right sided systems usually. However, convergence tells you the stability of a system. A lot of systems have feedback, which changes stability. Convergence and regions of stability are very useful.

Interpreting the Z transform

Similar to the Laplace transform, we can write a rational polynomial:

$$\begin{aligned} X(z) &= \frac{a_0 z^{-m} + a_1 z^{-(m-1)} + \dots + a_m}{b_0 z^{-n} + b_1 z^{-(n-1)} + \dots + b_n} \\ &= k \frac{\overbrace{\prod_{i=1}^m (1 - c_i z^{-1})}^{\text{zeros } c_i}}{\underbrace{\prod_{k=1}^n (1 - p_k z^{-1})}_{\text{poles } p_k}} \end{aligned}$$

We can't have convergence anywhere that includes a pole, so we can make similar figures as from the Laplace transform.

For example:

$$X(z) = \frac{1 + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 + z^{-1})(1 - 2z^{-1})}$$

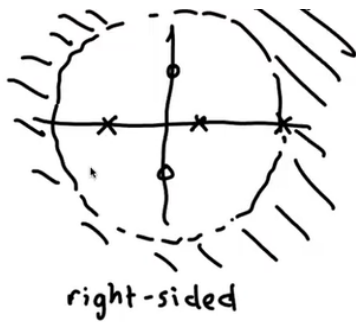
There are three poles here:

$\frac{1}{2}$, -1 , 2 . There are zeros at $\pm j$.

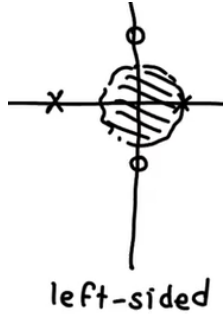
We know that the expansion will be something of the form

$$\frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - z^{-1}} + \frac{C}{1 - 2z^{-1}}$$

If we sum all three terms, we could have a right sided signal with the region of convergence outside the circle (all things at radius beyond the furthest pole, here at 2).



If we use the innermost radius, we get a left sided signal



Or in between radius 1 and 2 (two sided)



Where the A and B term are right sided and the C term is left sided.

The last option is between the innermost circle and the circle in the middle, which is also two sided. This has the A as right sided, and the B and C terms as left sided.

Z transform and the DTFT

By inspection we see a strong similarity between the $\mathbb{Z}$ transform and the DTFT.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

The $DTFT$ is the z transform evaluated at $z = e^{j\omega}$, so the DTFT is a special case of the $\mathbb{Z}$ transform, or the $\mathbb{Z}$ as a generalization of the DTFT.

Graphically, we can evaluate the DTFT from the pole zero plot.

For rational

$$X(z) = k \frac{\prod_{i=1}^m (z - c_i)}{\prod_{k=1}^n (z - p_k)},$$

$$X(e^{j\omega}) = |k| \frac{\prod_{i=1}^m |e^{j\omega} - c_i|}{\prod_{k=1}^n |e^{j\omega} - p_k|}$$

Graphically, let's think about each term. In the top, this is the distance from a point on the unit circle to the point c_i . Similarly, the denominator is the distance to each of the poles.

$$\angle X(e^{j\omega}) = \angle k + \sum_{i=1}^m \angle(e^{j\omega} - c_i) - \sum_{k=1}^n \angle(e^{j\omega} - p_k)$$

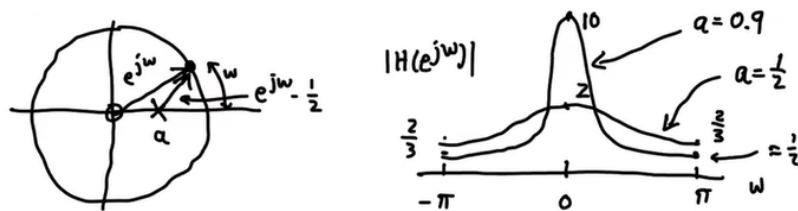
This tells us graphically why the magnitude and phase response is how it is.

Example: $H(z) = \frac{1}{1-az^{-1}}, |z| > |a|$

$$H(z) = \frac{z}{z-a}$$

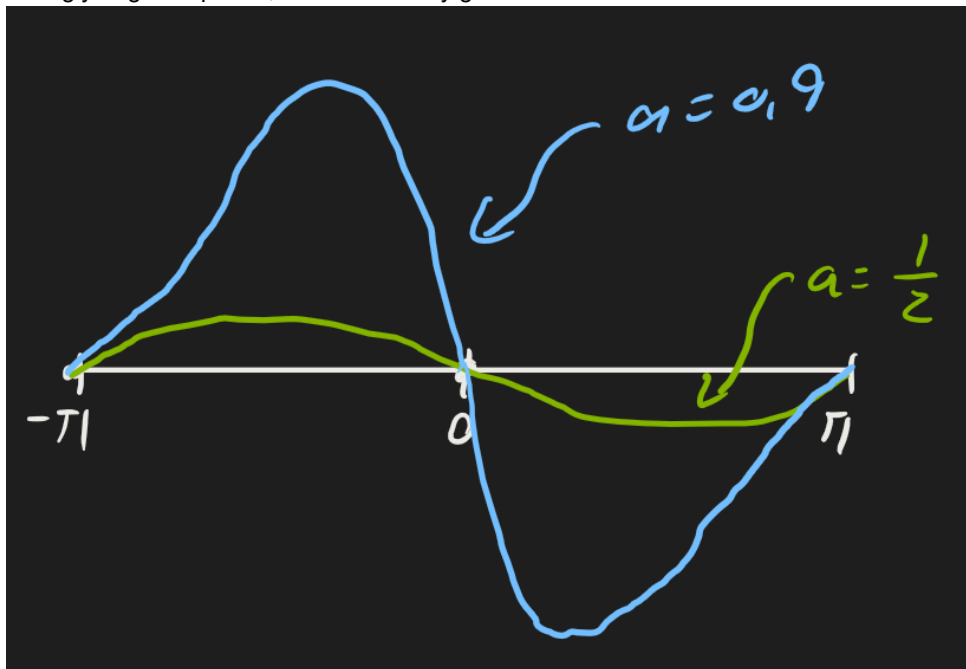
$$H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a}$$

The magnitude of the numerator is always 1, so we're just spinning around the complex plane in a circle. In the denominator, as we spin we are getting further and closer to a , so we have a cyclic magnitude that looks generally like a sin wave but with its own min and max.



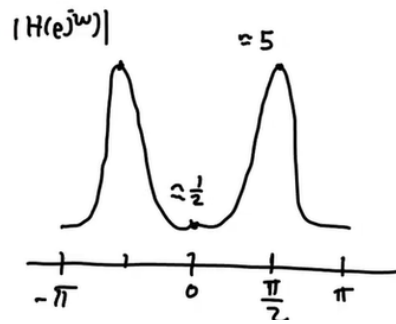
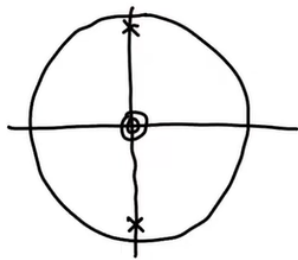
As we change the argument a , we change how stark the difference is proportionally throughout the traversal of the circle.

We can also graphically see the phase (we are comparing the angle between the x axis, the origin, and the point minus the angle from the x axis, the pole a , and the point). If the pole a is close to 1, then it will hit 90 degrees very quickly and make a strongly negative phase, and then slowly get to 180.



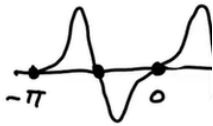
Example: $H(z) = \frac{z^2}{(z+0.9j)(z-0.9j)}, |z| > 0.9$

We can get the rough shape for the magnitude here. The distance between the zeros is 1, and the distance to the poles is close to $\sqrt{2}$, and we have two so $\sim \sqrt{2}\sqrt{2} \approx 2$, so $\frac{1}{2}$.



The poles are symmetric about zero, so the phase at 0 is 0. at $\frac{\pi}{2}$ its 0, and the same at π - the angles between each of the poles cancel out. The phase is symmetric, and will be very sharp:

$$\angle H(e^{jw})$$



Z transform and LTI systems

A convolution in n (time domain) is a product in the $\mathbb{Z}$.

$$\begin{array}{lcl} x[n] \rightarrow \boxed{h[n]} \rightarrow y[n] = x[n] * h[n] & \uparrow \text{time} \\ X(z) \rightarrow \boxed{H(z)} \rightarrow Y(z) = X(z) \cdot H(z) & \downarrow \mathbb{Z} \\ & \uparrow \\ & \text{system function or transfer function} \end{array}$$

If $H(z)$ is causal, the region of convergence is the exterior of a circle, including infinity.

Causality

If $H(z)$ is rational, then this is equivalent to saying that the ROC is outside the outermost pole, and the order of the numerator is $\leq$ the order of the denominator.

For example,

$$H(z) = \frac{z}{z^2 - 1} \text{ has } \infty \in ROC$$

but

$$H(z) = \frac{z^3}{z^2 - 1} \text{ has } \infty \notin ROC$$

Stability

$H(z)$ is stable $\Leftrightarrow$ ROC includes the unit circle.

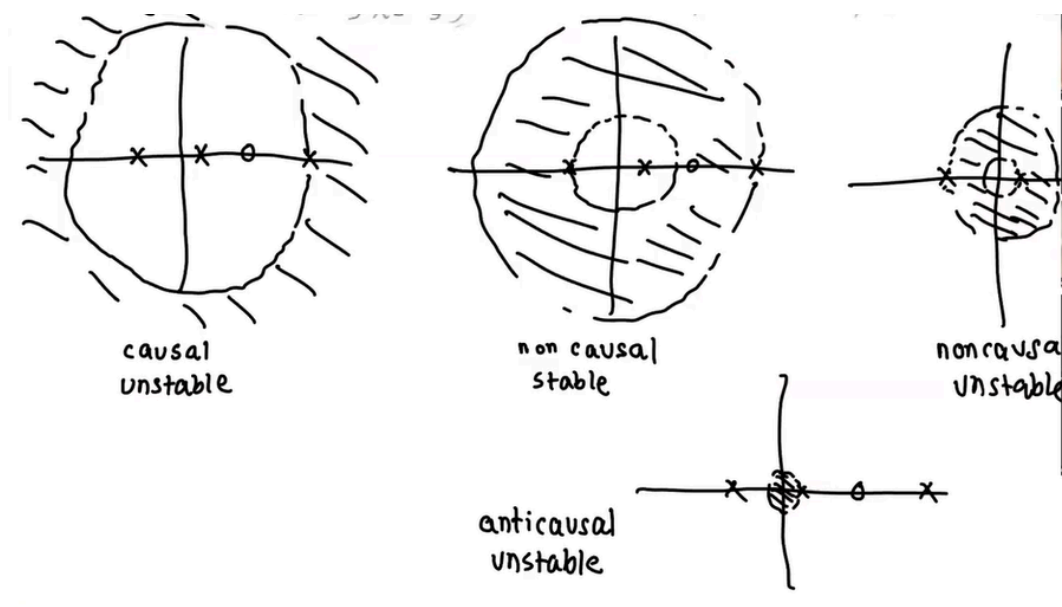
If it is causal, rational H_z is stable $\Leftrightarrow$ all poles are inside the unit circle.

Example: $H(z) = \frac{z-1}{(z-2)(z+\frac{2}{3})(z-\frac{1}{3})}$

We have three poles, at $2, -\frac{2}{3}, \frac{1}{3}$.

If we have everywhere outside of the outermost, this is causal but unstable.

If we have between $\frac{2}{3}$ and 2 , it is noncausal but stable. If we have between the innermost $-\frac{1}{3}$ and $\frac{2}{3}$ then it is noncausal AND unstable (yikes).



When to $\mathbb{Z}$ transform?

Whenever we are dealing with frequencies, we want to use Fourier. The Z transform is useful when you want to find the step response, stability, or Linear Constant Coefficient Difference Equations.

Lets calculate the step response from a given system response.

* Step response

Example: Calculate step response of system with $h[n] = (\frac{1}{2})^n u[n]$

$$x[n] = u[n] \xleftrightarrow{\mathbb{Z}} X(z) = \frac{1}{1-z^{-1}}, |z| > 1$$

$$h[n] = (\frac{1}{2})^n u[n] \xleftrightarrow{\mathbb{Z}} H(z) = \frac{1}{1-\frac{1}{2}z^{-1}}, |z| > \frac{1}{2}$$

$$\begin{aligned} Y[n] = x[n] * h[n] &\xleftrightarrow{\mathbb{Z}} Y(z) = X(z)H(z) \\ &= \frac{1}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} \\ &= \frac{2}{1-z^{-1}} - \frac{1}{1-\frac{1}{2}z^{-1}}, |z| > 1 \end{aligned}$$

$$y[n] = 2u[n] - (\frac{1}{2})^n u[n]$$

From table,

$$a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}}, |z| > |a|$$

$$-a^n u[-n-1] \leftrightarrow \frac{1}{1-az^{-1}}, |z| < |a|$$